Prompt Engineering

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The problem

- A chatbot that assists engineers with designing distribution and transmission grids should be able to solve practice problems from a professional engineering exam.
- To experiment with prompt engineering, I have an LLM (Mistral 7B), Claude, and Llama-3-3b attempt to answer three questions taken from a Power Exam Sample Test
- It should be noted that this sample test is available as a free-access PDF on the internet and therefore was likely included in the model's training data. Never-before seen problems may experience worse performances.

1. A power supply serves a 40 HP, 480V motor at 85% power factor and a 10 HP, 480V motor at 95% power factor. The motors are three phase induction motors. What size power factor correction capacitors must be provided to the power supply in order to achieve a total power factor of 95%?
   - **Answer: 8.9 KVAR (102.47 uF)**
2. Which of the following is most likely not a primary purpose of bonding or electrically connecting two objects together?
   - (A) Minimize the voltage difference between the two objects.
   - (B) Provide a connection to earth (ground).
   - (C) Prevent sparking between two objects.
   - **(D) Allow high currents to pass through the objects with minimal arcing.**
3. Which of the following locations is most appropriate location for a surge protection device?
   - (A) In series at the point of use of a critical load.
   - **(B) In parallel with the power supply service entrance to a computer server facility.**
   - (C) On a conductor prior to a VFD, in order to limit harmonics.
   - (D) On a conductor that serves equipment with low power factors.

The prompts

- Zero-shot
  - Just the question, no prompt template
- Role prompt
  - Provide the chatbot with a role and then ask the question
  - *You are assisting an electrical engineer responsible for designing distribution and transmission lines. Here is the question: <question>*
- Decomposition prompt
  - Ask the chatbot to deconstruct the problem before providing a solution
  - *Determine if the question is a math problem or a reasoning question. Next, if it is a math problem, state the variables, how they relate, and how you would solve the problem. If the question is not a math problem, restate the important information provided in the question and any additional electrical engineering information that may be relevant. Then solve the problem and provide the answer. Here is the question: <question>*
- Step-by-step prompt
  - Instruct the chatbot to think through each step and then give an answer
  - *Determine if the question is a math problem or a reasoning question. If it is a math problem, write out the steps to solve it and state the final answer. If it is a reasoning problem, consider the possible answers and provide justification for whether it is correct. Here is the question: <question>*
- Full prompt
  - Combine all the previous prompt strategies into a single prompt
  - *You are assisting an electrical engineer responsible for designing distribution and transmission lines. First, determine if the question is a math problem or a reasoning question. Next, if it is a math problem, state the variables, how they relate, and how you would solve the problem. If the question is not a math problem, restate the important information provided in the question and any additional electrical engineering information that may be relevant. If it is a math problem, write out the steps to solve it and state the final answer. If it is a reasoning problem, consider the possible answers and provide justification for whether it is correct. End by stating the final answer with units if necessary. Here is the question: <question>*

The answers

Model	Parameters	Q	Correct Answer	Zero Shot	Role	Deconstruction	Step-by-Step	Full Prompt
Claude Sonnet	175 billion	1	8.9 kvar or 102.47 uF	8.66 kvar	8.68 kvar	8.65 kvar	8.7 kvar	8.68 kvar
		2	D	B	B	D	D	D
		3	B	B	B	B	B	B
Mistral instruct	7 billion	1	8.9 kvar or 102.47 uF	55 uF	7.4 uF	No answer	160 uF	2.98 kvar
		2	D	A	D	D	B	D
		3	B	D	D	A	A	B
Llama 3 instruct	3 billion	1	8.9 kvar or 102.47 uF	5.3 mF	3.3-10 uF	33F	$\boxed{2671}$	17 mF
		2	D	D	D	D	D	D
		3	B	B	B	B	B	B

Notes

- Claude did the best with the math questions, but gave a different answer each time.
- Mistral did not do well with step-by-step prompting
- The full prompt and deconstruction seemed the most effective across the board
- More complex prompting led to longer outputs
  - But not better math
- Llama-3-3B-instruct at one point output "`The question is from the 2006 IEEE exam.`" indicating that it had seen the answers before in the training data.
- Claude output "let's think this through step by step" without prompting
- Llama-3-3B instruct said every question was a reasoning question even if it was a math question
- 58 pages of total output copied

Output Examples

- **Llama-3-3b-instruct Question 3 Full prompt - Correct**

- This question is a reasoning question. Bonding or electrically connecting two objects together serves several primary purposes in electrical engineering. Minimizing the voltage difference between two objects (A) is actually a secondary effect of bonding, as it helps to prevent voltage differences from causing arcing or sparking. Providing a connection to earth (ground) (B) is a primary purpose of bonding, as it ensures that all metal parts of a system are connected to the earth, preventing electrical shock and short circuits. Preventing sparking between two objects (C) is also a primary purpose of bonding, as it helps to prevent arcing and electrical discharges between objects that are not intended to be in electrical contact. Allowing high currents to pass through the objects with minimal arcing (D) is not a primary purpose of bonding. In fact, bonding is intended to prevent arcing, not allow it. Therefore, the correct answer is (D) Allow high currents to pass through the objects with minimal arcing. This is because bonding is primarily used to ensure safety, prevent electrical shock, and prevent arcing, rather than to facilitate the passage of high currents.

- More output examples can be provided in a word document, but most text is too long to fit on a given slide.

Mistral-7b-instruct Question 1 Full prompt - Incorrect

- This question is a reasoning problem, not a math problem. The important information provided in the question is that we have two motors with different power factors (85% for the 40 HP motor and 95% for the 10 HP motor), both operating at 480V, and we need to find the size of power factor correction capacitors required to achieve a total power factor of 95%. To solve this problem, we first need to calculate the reactive power (Q) consumed by each motor. The active power (P) and reactive power (Q) are related through the power factor (pf) according to the formula:
- P = (Pf * Pactive)
- Q = (Sqrt(1 - (Pf^2))) * Pactive
- where Pactive is the active power in watts and S is the apparent power in VA (Volts * Amperes). Since the motors are three-phase, the total active power (Ptotal) and reactive power (Qtotal) are the sum of the active and reactive power for each motor. Let's denote:
- - P40 as the active power of the 40 HP motor (40 HP * 746 W/HP = 30,040 W)
- P10 as the active power of the 10 HP motor (10 HP * 746 W/HP = 7,460 W)
- - Q40 as the reactive power of the 40 HP motor
- - Q10 as the reactive power of the 10 HP motor
- - Qtotal as the total reactive power of both motors
- - Qcorr as the reactive power that needs to be corrected to achieve a total power factor of 95%

- We can write the following equations:

- Ptotal = P40 + P10 = 30,040 W + 7,460 W = 37,500 W
- Qtotal = Q40 + Q10
- Qcorr = (Sqrt(1 - (0.95^2))) * Ptotal

- Since we don't have the reactive power values for each motor, we can't calculate the exact size of the power factor correction capacitors. However, we can find the reactive power that needs to be corrected to achieve a total power factor of 95%. To achieve a total power factor of 95%, the reactive power correction should be:

- Qcorr = (Sqrt(1 - (0.95^2))) * Ptotal = (Sqrt(1 - 0.9025)) * 37,500 W
- Qcorr ≈ 2,985.73 W

Since the power factor correction capacitors store and release reactive power in units of VAR (Volt-Amperes Reactive), the size of the power factor correction capacitors required to achieve a total power factor of 95% is approximately 2,985.73 VAR. Final answer: Approximately 2,985.73 VAR of power factor correction capacitors are required to achieve a total power factor of 95%